



1. For how many integers n does $2^{2^n} = 2^{n^2}$ hold?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

2. Jerry is at the gym and is going to use the bench press. He wants to put on 5, 5, 10, 10, 10, and 20 pound weights on either side of the bar such that the bar is balanced. If equal weights are indistinguishable, the two sides of the bar are distinguishable, he must use all six weights, and order doesn't matter, how many ways can he put on these weights?

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 8

3. Let $f(n) = n^2 + n + 2020$. What is the sum of the distinct prime factors of the number $f(2019)$?

- (A) 90 (B) 108 (C) 110 (D) 111 (E) 676

4. Let n be a positive integer, and consider the set $S = \{1, 4, 9, 16, 25, 36, n\}$ where all seven of its elements are distinct. We replace every odd element k of S with $k + 1$, and we replace every even element k of S with $k - 1$. It is given that the median of the resulting set is equal to the median of S . How many possible values of n are there?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 6

5. Let $a_0, a_1, \dots, a_{2019}$ be real numbers such that $a_0 = 0$ and $a_n + a_{n+1}^{-1} = 2$ for every integer $n \geq 0$. What is the value of the product $a_1 a_2 \cdots a_{2019}$?

- (A) $\frac{1}{2020}$ (B) $\frac{1}{2019}$ (C) $\frac{1}{1010}$ (D) $\frac{1}{1009}$ (E) 1

6. Tasty and Stacy are playing a game. Tasty rolls a fair 6-sided die and Stacy rolls a fair n -sided die, whose faces are numbered from 1 to n . Stacy wins if the sum of the values of the two dice is divisible by 5. For which of the following values of n does Stacy maximize her chances to win?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

7. How many subsets S of the following four are there such that it is possible for all statements in S to be true and all other statements to be false?

At least one of these statements is true.

Both this statement and the following statement are true.

This statement is true.

The first statement is false.

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

8. Let $i = \sqrt{-1}$. What is the largest positive integer n such that the equation $(z - 1)(\bar{z} + 1) = 2019 + ni$ has at least one solution $z \in C$?

- (A) 89 (B) 90 (C) 91 (D) 92 (E) 93

9. Positive integers b_4 and b_6 have the property that there is a unique positive integer b_5 such that $1 > \frac{b_4}{4^2} > \frac{b_5}{5^2} > \frac{b_6}{6^2}$. If $b_4 + b_6 = 24$, what is the unique value of b_5 ?

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

10. There exists a positive integer n such that $\sqrt{n + \sqrt{n + \sqrt{n + \cdots}}} = \frac{1000}{n + \frac{1000}{n + \frac{1000}{n + \cdots}}}$. What is the sum of the digits of n ?

- (A) 1 (B) 2 (C) 9 (D) 10 (E) 18

11. Let $ABCD$ be a rectangle. Suppose the vertices of $ABCD$ have x -coordinates of 1, 2, 3, 4 in some order. What is the minimum possible area of $ABCD$?

- (A) 2 (B) 4 (C) $\frac{9}{2}$ (D) 8 (E) 9

12. Consider a semicircle Γ with diameter AB . Consider the set S of circles tangent to both Γ and $\overline{AB}$. The centers of all circles in S form a locus that is a portion of a

- (A) hyperbola (B) non-circular ellipse (C) circle (D) parabola (E) none of these

13. For a set of positive integers S , its *primality* is the number of primes that divide some element of S . For instance, $\{1\}$ has primality 0 and $\{12, 21\}$ has primality 3. If a set S of n consecutive integers has primality 10, what is the maximum possible value of n ?

- (A) 1 (B) 10 (C) 16 (D) 30 (E) 46

14. Let m be the number of digits in $9!^{9!}$, and let n be the number of digits in $10!^{10!}$. Which of the following values is closest to $\frac{m}{n}$?

- (A) $\frac{1}{14}$ (B) $\frac{1}{13}$ (C) $\frac{1}{12}$ (D) $\frac{1}{11}$ (E) $\frac{1}{10}$

15. Let ABC be a triangle with $AB = 30$, $BC = 51$, $CA = 63$. Points P and Q lie on $\overline{BC}$, R lies on $\overline{CA}$, and S lies on $\overline{AB}$ such that $PQRS$ is a parallelogram, and the center of $PQRS$ coincides with the centroid of $\triangle ABC$. What is the area of parallelogram $PQRS$?

- (A) 84 (B) 126 (C) 168 (D) 336 (E) 378

16. Over all real numbers $N > 0$, what is the maximum possible number of permutations a, b, c, d, e of the integers 2, 3, 5, 7, 11 such that it is possible to put parentheses around the expression $a \div b \div c \div d \div e$ so that it equals N ?

- (A) 6 (B) 24 (C) 36 (D) 60 (E) 120

17. Let a, b, c, d be distinct positive real numbers forming a geometric progression, in that order. Suppose that $\log a, \log b, \log \gamma, \log d$ form a geometric progression too, in that order. If $a = \sqrt{2}$, what is γ ?

- (A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4 (E) $4\sqrt{2}$

18. The value of $\tan\left(\frac{\pi}{24}\right) + \tan\left(\frac{3\pi}{24}\right) + \tan\left(\frac{5\pi}{24}\right) + \tan\left(\frac{7\pi}{24}\right) + \tan\left(\frac{9\pi}{24}\right) + \tan\left(\frac{11\pi}{24}\right)$ can be expressed as $\sqrt{m} + \sqrt{n}$, where m and n are integers. What is $m + n$?

(A) 56 (B) 98 (C) 104 (D) 112 (E) 200

19. Let a and b be positive real numbers satisfying $(\log_2 a)(\log_4 b) + a^2 = 135$ and $(\log_4 a)(\log_2 b) + b^2 = 263$. Then the value of ab can be written as $p\sqrt{q}$ for positive integers p and q such that q is not divisible by the square of any prime. What is $p + q$?

(A) 66 (B) 130 (C) 258 (D) 514 (E) 1026

20. Rectangle $ABCD$ is a golden rectangle, meaning that if we construct points P_1 and P_2 on $\overline{AB}$ and $\overline{CD}$, respectively, such that AP_1P_2D is a square, then rectangles $ABCD$ and P_2P_1BC are similar. In other words, P_2P_1BC is also a golden rectangle. Continue this construction on rectangle P_2P_1BC : construct points P_3 and P_4 on $\overline{P_2P_1}$ and $\overline{BC}$, respectively, such that $P_2P_3P_4C$ is a square, so that $P_4P_3P_1B$ is also a golden rectangle. If this construction repeats infinitely, then there is one point G inside all of these golden rectangles. What is $\tan \angle BAG$?

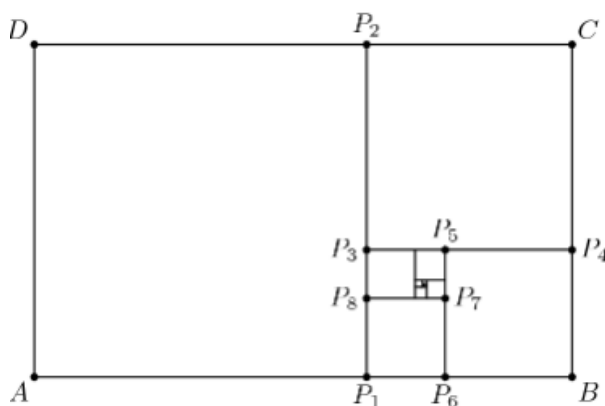


Figure 1: Problem 20

(A) $\frac{\sqrt{5}-1}{2}$ (B) $\sqrt{5} - 1$ (C) $\sqrt{5} - 2$ (D) $2 - \sqrt{3}$ (E) $3 - \sqrt{5}$

21. How many ordered triples (a, b, c) of not necessarily distinct positive integer divisors of 216,000 satisfy $\text{lcm}(a, \text{gcd}(b, c)) = \text{lcm}(b, \text{gcd}(a, c)) = \text{lcm}(c, \text{gcd}(a, b))$?

(A) 33,880 (B) 81,200 (C) 106,400 (D) 120,960 (E) 212,800

22. In triangle ABC , let D, E, F denote the feet of the altitudes from A, B, C , respectively. If the distance from A to $\overline{EF}$ is 2, the distance from B to $\overline{FD}$ is 3 and the distance from C to $\overline{DE}$ is 6, what is the area of $\triangle DEF$?

(A) 3 (B) $3\sqrt{3}$ (C) 6 (D) $6\sqrt{3}$ (E) 9

23. There exists $\triangle ABC$ with $\angle B = 30^\circ$ that satisfies $\frac{b+c}{2\cos C} = a$. Find $\angle A$. (Here, $a = BC$, $b = CA$, $c = AB$.)

(A) 100 (B) 112.5 (C) 135 (D) 140 (E) 160

24. Call a four-digit positive integer *asuboptimal* if the sum of two of its digits is equal to the sum of the other two digits. For instance, 1234, 2020, and 9801 are all asuboptimal. How many four-digit positive integers are asuboptimal?

(A) 1494 (B) 1557 (C) 1584 (D) 1632 (E) 1845

25. Feynman is stuck in a bunker with only a clock to tell time; however, the minute and hour hands have the same length, so it is impossible to tell them apart. Suppose that Feynman is able to accurately predict the time to an error of at most 3 hours. If at one instant he looks at the clock but determines the time incorrectly, which of these is nearest to the maximum possible error in minutes from the actual time?

(A) 165 (B) 166 (C) 167 (D) 168 (E) 169

Time limit: 75 minutes